

Singularity of finitely supported random walks on Fuchsian groups

A mathematical exposition of the Lean formalization

24 September 2026

Abstract

We present the argument implemented in the Lean development for singularity of the hitting measure of a finitely supported, semigroup-generating random walk on any discrete nonelementary subgroup of $\mathrm{PSL}_2(\mathbb{R})$. No symmetry or finite-covolume assumption is made. The cocompact argument combines a one-sided comparison of invariant currents with an infinite-strip Green factorization and a Fourier obstruction. For the general case, an ideal end of a Dirichlet cell supplies a bounded-height endpoint. Boundary minimality then gives finite separators. Exact entrance identities yield common shadow estimates, and an explicit Radon–Nikodym rigidity argument forces orbit quasiconvexity under hypothetical nonsingularity. This reduces to the cocompact case or a visual-null limit set. The relevant Lean declarations are indexed throughout and in an appendix.

Contents

1	Statement, conventions, and the proof’s dependencies	2
2	The walk, Green functions, and boundary densities	2
3	Green compression and exact separator identities	4
4	The cocompact boundary package	5
5	Why one-sided absolute continuity suffices	6
6	The periodic strip and its boundary factorization	7
7	The Fourier obstruction	8
8	Quasiconvex orbits and a Green-distance criterion	10
9	Dirichlet ends and bounded-height endpoints	10
10	Finite separators and harmonic deficits	11
11	Common shadows for harmonic and visual measures	13
12	From nonsingularity to a uniform Green comparison	14
13	Completion of the proof	16
A	How to locate the proof in Lean	16
B	Scope of this exposition and classical inputs	17

1 Statement, conventions, and the proof's dependencies

Theorem 1.1. *Let $\Gamma < \mathrm{PSL}_2(\mathbb{R})$ be a discrete nonelementary subgroup. Let μ be a probability measure with finite support S , where S generates Γ as a semigroup. For independent μ -distributed increments $X_1, X_2, \dots$, put $Z_n = X_1 \cdots X_n$. Then $Z_n o$ converges almost surely to $\partial\mathbb{H}^2$, and its limiting distribution ν is mutually singular with visual measure:*

$$\nu \perp m_o.$$

This holds for every $o \in \mathbb{H}^2$, without symmetry of μ and without a restriction on the covolume of Γ .

LEAN: `FuchsianSingularity.fuchsian_hittingMeasure_singular`.

Here and below a Lean reference of the form `Module.declaration` means the declaration in `Singularity/Module.lean`; its namespace is `Singularity`. The stated scope is finite support with *semigroup* generation. Group generation alone is not substituted for it. Nonelementarity means that no point of $\mathbb{H}^2 \cup \partial\mathbb{H}^2$ has a finite orbit. The upper half-plane has curvature -1 , and $\partial\mathbb{H}^2 = \mathbb{R} \cup \{\infty\}$. At $z = x + iy$ the visual probability is

$$dm_z(t) = \frac{y}{\pi((t-x)^2 + y^2)} dt, \quad m_z\{\infty\} = 0. \quad (1.1)$$

All m_z are equivalent. When $o = i$, its relative Poisson kernel is

$$P_z(t) = \frac{dm_z(t)}{dm_i} = \frac{y(1+t^2)}{(t-x)^2 + y^2}. \quad (1.2)$$

We use the Busemann sign convention

$$\beta_\xi(z, w) = \lim_{u \rightarrow \xi} (d(z, u) - d(w, u)), \quad P_z(\xi) = e^{\beta_\xi(i, z)}.$$

In particular $\beta_\infty(z, w) = \log(\mathrm{Im} w / \mathrm{Im} z)$.

Relation to the formal proof. This is a mathematical exposition, not a literal translation of tactic scripts. It gives the main estimates and their assembly, including the noncompact argument. We use standard foundational theorems of probability, measure theory, Fourier analysis, and hyperbolic geometry in their usual mathematical form. The cocompact Martin–Naïm boundary package is stated explicitly in Proposition 4.1; it is a substantial standard theorem, not something claimed to follow merely from weak convergence of hitting laws. The Lean development constructs the needed package and the other inputs internally; the appendix identifies its modules. This document does not rederive every foundational theorem or every one of the development's 2,591 declarations.

The proof is organized without circularity: first prove the cocompact case; then prove singularity for quasiconvex orbits; next establish the geometric separator property for noncompact groups; finally derive quasiconvexity from hypothetical nonsingularity. The final step does not assume a classification of finitely generated Fuchsian groups, relative hyperbolicity, or the singularity conclusion of an external rigidity theorem.

2 The walk, Green functions, and boundary densities

Write e for the identity, $p_n(x, y) = \mathbb{P}_x(Z_n = y)$, and

$$G(x, y) = \sum_{n \geq 0} p_n(x, y), \quad G_0 = G(e, e), \quad F(x, y) = \mathbb{P}_x(T_y < \infty) = \frac{G(x, y)}{G_0}. \quad (2.1)$$

The equality for F follows by restarting the walk at its first visit to y . Define

$$d_G(x, y) = -\log F(x, y), \quad D_G(g) = d_G(e, g), \quad D(g) = d(o, go).$$

The Green distance need not be symmetric. It is left invariant and satisfies the triangle inequality, since $F(x, z) \geq F(x, y)F(y, z)$.

Proposition 2.1 (Basic random-walk facts). *Under the hypotheses of Theorem 1.1:*

1. On counting $\ell^2(\Gamma)$,

$$(Pf)(x) = \sum_{s \in S} \mu(s)f(xs), \quad (P^*f)(x) = \sum_{s \in S} \mu(s)f(xs^{-1})$$

satisfy $\|P\| \leq 1$ and $r(P) < 1$. In particular $G = (I - P)^{-1}$ is bounded, the Green series converges in operator norm, and all $G(x, y)$ are positive.

2. The geometric boundary limit exists almost surely. Its law ν is stationary, atomless, supported on the ideal limit set Λ , and $g_*\nu \sim \nu$ for every $g \in \Gamma$.

3. If $E \subset \partial\mathbb{H}^2$ is measurable, then

$$(Z_n)_*\nu(E) = \nu(Z_n^{-1}E) \longrightarrow \mathbf{1}_E(Z_\infty) \quad \text{almost surely.} \quad (2.2)$$

4. Either $\nu \perp m_o$ or $\nu \ll m_o$.

Explanation of the inputs and their use. Jensen's inequality and invariance of counting measure give $\|Pf\|_2^2 \leq \sum_s \mu(s) \sum_x |f(xs)|^2 = \|f\|_2^2$. A nonelementary Fuchsian group contains a free subgroup by hyperbolic ping-pong and is nonamenable. The admissible convolution spectral-radius criterion gives $r(P) < 1$; see Woess [1], Chapter 12. This is a spectral-radius assertion and does *not* assert $\|P\| < 1$. The right and left convolution formulations are intertwined by $f(x) \mapsto f(x^{-1})$. The spectral-radius formula supplies constants $A < \infty$, $q < 1$ with $\|P^n\| \leq Aq^n$, proving Green convergence. Semigroup generation supplies a positive-probability path between any two states, giving positivity of G .

Almost-sure boundary convergence is the standard convergence theorem for nonelementary walks on $\mathbb{H}^2$; in the formal development it is obtained from the actual path law and the projective action. Stationarity follows by conditioning on the first increment. Iterating stationarity along positive words proves equivalence with every group translate. A maximal atom, if present, would give a nonempty finite invariant set of maximal atoms, which nonelementarity excludes. Discreteness and escape from finite sets place the geometric limit in Λ .

Given the first n increments, the remaining boundary limit has law $(Z_n)_*\nu$. Thus the left side of (2.2) is the conditional expectation of $\mathbf{1}_E(Z_\infty)$; martingale convergence proves the claim. For the last assertion, take a visual-null Borel set carrying the singular part of ν and replace it by the union of all its group translates. The resulting set B is invariant and visual-null, and $\nu(B)$ is exactly the mass of the singular part. The event $Z_\infty \in B$ is invariant under the one-sided shift of the independent increment sequence: deleting the first increment changes the boundary limit by a group element. Ergodicity of the independent shift gives $\nu(B) \in \{0, 1\}$. These alternatives give respectively $\nu \ll m_o$ and $\nu \perp m_o$. $\square$

LEAN: FuchsianHittingMeasure; FuchsianBoundaryConcentration; FuchsianHittingErgodicity.

All almost-everywhere identities involving elements of Γ can be imposed on one invariant conull set: Γ is countable and its action is quasi-invariant. This convention is used throughout.

For any finite quasi-invariant measure ρ , put

$$K_\rho(g, \xi) = \frac{d(g_*\rho)}{d\rho}(\xi), \quad \sigma_\rho(g, \xi) = -\log K_\rho(g^{-1}, \xi). \quad (2.3)$$

Then

$$\sigma_\rho(gh, \xi) = \sigma_\rho(g, h\xi) + \sigma_\rho(h, \xi), \quad (2.4)$$

$$\rho(gE) = \int_E e^{-\sigma_\rho(g, \xi)} d\rho(\xi). \quad (2.5)$$

If $\nu = f\rho$ and $H = \log f$, then

$$\sigma_\nu(g, \xi) - \sigma_\rho(g, \xi) = H(\xi) - H(g\xi). \quad (2.6)$$

For ν , $g \mapsto K_\nu(g, \xi)$ is positive μ -harmonic and takes value one at e . Optional stopping at a singleton gives, for any positive normalized superharmonic h ,

$$F(x, y)h(y) \leq h(x). \quad (2.7)$$

Applying this in both directions to $K_\nu(\cdot, \xi)$ yields

$$-d_G(g, e) \leq \sigma_\nu(g, \xi) \leq D_G(g). \quad (2.8)$$

The visual counterpart is $|\sigma_{m_o}(g, \xi)| \leq D(g)$.

LEAN: `StationaryLogCocycle`; `RadonNikodymCoboundary`; `VisualLogCocycle`.

3 Green compression and exact separator identities

These facts apply to any of the groups in the theorem.

Lemma 3.1 (Coercivity without laziness). *For $v \in \ell^2(\Gamma)$,*

$$\operatorname{Re}\langle Gv, v \rangle \geq \frac{1}{2}\|v\|_2^2. \quad (3.1)$$

For every $A \subset \Gamma$, let $i_A : \ell^2(A) \rightarrow \ell^2(\Gamma)$ be extension by zero, and let $r_A = i_A^$ be restriction. Then $G_A = r_A G i_A$ is boundedly invertible. Writing $M_A = G_A^{-1}$, we have $\|M_A\| \leq 2$.*

Proof. Set $w = Gv$, so $v = w - Pw$. Direct expansion gives

$$2\operatorname{Re}\langle Gv, v \rangle - \|v\|^2 = \|w\|^2 - \|Pw\|^2 \geq 0.$$

Extension by zero transfers this inequality to G_A . Cauchy–Schwarz gives $\|G_A u\| \geq \|u\|/2$, so its range is closed. Its adjoint satisfies the same real-part inequality, hence $\ker G_A^* = \{0\}$; the range is therefore dense, and thus all of $\ell^2(A)$. The inverse norm bound follows. No self-adjointness or entrywise positivity of M_A is asserted. $\square$

LEAN: `Green`; `GreenAdjoint`; `Operators`.

For $T_A = \inf\{n \geq 0 : Z_n \in A\}$ define

$$F_A(x, a) = \mathbb{P}_x(T_A < \infty, Z_{T_A} = a), \quad a \in A.$$

This counts paths up to their *first* entrance into A ; by contrast, $G(x, a)$ counts all visits and places no restriction on intermediate paths. The first-entrance decomposition gives

$$G(x, y) = G^A(x, y) + \sum_{a \in A} F_A(x, a)G(a, y), \quad (3.2)$$

where G^A counts paths killed at entrance into A . Since $0 \leq F_A(x, a) \leq G(x, a)$, positivity implies a bounded entrance operator: $|F_A u| \leq G i_A |u|$. Restricting (3.2) to $y \in A$ gives $G i_A = F_A G_A$, hence $F_A = G i_A M_A$. Consequently, if every supported path from x to y meets A ,

$$G(x, y) = G(x, A) M_A G(A, y). \quad (3.3)$$

Rows and columns of G belong to ℓ^2 because G is bounded. Thus the right side is a legitimate Hilbert-space pairing. A two-index matrix sum should be interpreted through finite-coordinate truncations converging in that pairing; boundedness of M_A alone does not imply absolute summability of its entrywise expansion. The equivalent Schur identity is

$$G_{BB}^A = G_{BB} - G_{BA} G_A^{-1} G_{AB}, \quad B = \Gamma \setminus A.$$

LEAN: EntranceOperator; GreenSeparator; FiniteEntranceDecomposition.

4 The cocompact boundary package

In this section assume $\Gamma \backslash \mathbb{H}^2$ is compact. Put $\check{\mu}(g) = \mu(g^{-1})$ and let $\check{\nu}$ be its hitting law. Then $\check{G}(x, y) = G(y, x)$.

Proposition 4.1 (Martin kernels and the invariant current). *The geometric circle is the Martin boundary for each of the two walks. There are positive continuous kernels*

$$K(g, \eta) = \lim_{y \rightarrow \eta} \frac{G(g, y)}{G(e, y)}, \quad \check{K}(g, \xi) = \lim_{x \rightarrow \xi} \frac{G(x, g)}{G(x, e)},$$

which agree almost everywhere with the densities of $g_* \nu$ and $g_* \check{\nu}$. Off the diagonal the limit

$$\Theta(\xi, \eta) = \lim_{x \rightarrow \xi, y \rightarrow \eta} \frac{G(x, y)}{G(x, e) G(e, y)} \quad (4.1)$$

exists, is positive and continuous, and

$$dJ(\xi, \eta) = \Theta(\xi, \eta) d\check{\nu}(\xi) d\nu(\eta) \quad (4.2)$$

is Γ -invariant. The associated flow measure $dJ dt$ descends to a finite, nonzero invariant measure on the compact frame quotient.

This is the finite-range cocompact Martin–Ancona–Naïm package. The order of the two measures matters: the first is reflected, the second forward. Background for the boundary results and current convention is provided by [2, 3]. The formalization proves the required Ancona comparison, Martin identification, off-diagonal convergence and covariance rather than assuming this proposition.

For clarity, the mechanism behind the package is as follows. Cocompactness gives a uniform orbit net. The spectral bound and exponential hyperbolic detours make the total weight of paths avoiding a large ball on a geodesic superexponentially small in the ball radius. First/last-entrance decompositions, iterated through a summable sequence of relative errors, give uniform Ancona inequalities

$$G(x, y) \asymp G(x, u) G(u, y) \quad \text{when } uo \text{ stays a bounded distance from } [xo, yo].$$

Comparing normalized positive harmonic functions on nested barriers gives unique Martin limits; integral representations identify them with actual hitting densities. Off-diagonal strip pairings give the two-variable limit. Its change-of-basepoint rule is exactly the invariance

in (4.2). Positivity and continuity imply local finiteness of $J dt$ in flow boxes; a compact fundamental cover gives finite mass on the quotient. These construction statements are the formal results indexed below, not extra hypotheses imposed on the walk.

LEAN: CocompactAxisAncona; GeometricMartinHomeomorph; HittingMartinDensity.

LEAN: GeometricNaimKernel; NaimCovariance; GeometricNaimQuotient.

5 Why one-sided absolute continuity suffices

Lemma 5.1. *In the cocompact case, $\nu \ll m_o$ implies that the normalized quotient measure induced by $J dt$ is Liouville probability. In particular, $\check{\nu} \ll m_o$ and $J = cJ_0$, where*

$$dJ_0(\xi, \eta) = \frac{d\xi d\eta}{(\xi - \eta)^2}, \quad c > 0. \quad (5.1)$$

Proof. Let L be Liouville probability on the compact frame quotient and let Ω be the normalized probability induced by $J dt$. Both are geodesic-flow invariant. The geodesic flow is ergodic for L by the Hopf argument (in the formalization, the corresponding Haar assertion follows from Mautner's argument and the contracting upper/lower unipotent groups).

Fix a continuous observable f . For L -almost every frame its forward Cesàro averages converge to $\int f dL$. This convergence property is unchanged along a strong stable leaf: the forward trajectories become asymptotic, and uniform continuity on the compact quotient makes the difference of their averages tend to zero. It is also unchanged by a fixed time shift.

In local Hopf coordinates (ξ^-, ξ^+, t) , Haar measure is equivalent to the product of visual measures in the two endpoint coordinates and Lebesgue measure in the time coordinate. Fubini, followed by stable saturation and time-shift invariance, says that for visual-almost every forward endpoint, the convergence property holds for every compatible backward endpoint and time. The possible diagonal has zero product mass. Because $\nu \ll m_o$, this full-measure assertion transfers to $\check{\nu} \otimes \nu \otimes dt$, even if $\check{\nu}$ is initially singular. The positive kernel Θ transfers it further to Ω .

Invariance of Ω makes the integral of each Cesàro average equal to $\int f d\Omega$. Bounded convergence gives $\int f d\Omega = \int f dL$. Continuous observables determine finite measures on the compact quotient, so $\Omega = L$. Lifting this equality to flow boxes and disintegrating in t gives $J = cJ_0$ on boundary pairs. Since Θ is positive, local product rectangles then imply $\check{\nu} \ll m_o$ (and equivalence of both marginals with m_o). $\square$

LEAN: OneSidedNaimRigidity.geometricNaimQuotientProbability_eq_haar_of_forward_ac.

LEAN: StableFrameMeasureTransfer; NaimQuotientComparison.

Lemma 5.2 (Uniform densities and Green comparison). *If ν is nonsingular in the cocompact case, there are positive constants a, b, C such that*

$$am_o \leq \nu \leq bm_o, \quad am_o \leq \check{\nu} \leq bm_o, \quad (5.2)$$

and, for all $x, y \in \Gamma$,

$$C^{-1}e^{-d(xo, yo)} \leq G(x, y) \leq Ce^{-d(xo, yo)}. \quad (5.3)$$

The same comparison holds for $\check{G}$.

Proof. Proposition 2.1 gives $\nu \ll m_o$, and Lemma 5.1 gives $J = cJ_0$. Write the two marginal densities as f_- and f_+ . On two separated compact boundary arcs U, V , the identity of currents reads $\Theta f_- f_+ = c dJ_0 / d(m_o \otimes m_o)$. Both continuous factors are bounded above and below by positive constants on $U \times V$. Integrating in one variable bounds the other marginal

density above and below on its arc. The integrals are positive and finite because the marginals are equivalent probabilities. A finite cover by such separated rectangles proves (5.2) globally.

First-hit domination gives $F(e, g)g_*\nu \leq \nu$. The density bounds imply $aF(e, g)m_{go} \leq bm_o$. The relative Poisson density has essential supremum $e^{D(g)}$, so $F(e, g) \leq (b/a)e^{-D(g)}$. For the reverse inequality, extend a geodesic through go to a boundary point η . Ancona comparison along an orbit sequence on that extension gives $G(e, g)K(g, \eta) \geq c_0 > 0$, with c_0 uniform. The density identification and (5.2) give $K(g, \eta) \leq (b/a)e^{D(g)}$. The almost-everywhere bound extends to every η by continuity and full support. Thus $G(e, g) \geq c_1e^{-D(g)}$. Left invariance gives (5.3), and transposition gives the reflected comparison. $\square$

LEAN: NaimCurrentRigidity; GreenVisualUpper; GreenVisualLower.

6 The periodic strip and its boundary factorization

Continue to assume that $\Gamma \backslash \mathbb{H}^2$ is compact and that $\nu \not\ll m_o$. Choose a hyperbolic element $a \in \Gamma$. After conjugating, its axis is the imaginary axis and $az = e^\tau z$, with $\tau > 0$. We may take $o = i$: changing the basepoint changes visual measure within its measure class and does not change the abstract walk. Let $L = \max_{s \in S} d(i, si)$. Choose a closed strip of width greater than L about the axis and let A consist of the group elements whose orbit points belong to that strip. Every walk path crossing from its negative side to its positive side meets A .

The strip modulo $\langle a \rangle$ is compact. Properness of the orbit map, including finite point stabilizers, therefore gives a disjoint parametrization

$$A = \{a^n b_j : n \in \mathbb{Z}, 1 \leq j \leq N\} \quad (6.1)$$

with $N < \infty$. Enlarge the width if necessary to ensure $N \geq 1$. The identification in (6.1) identifies $\ell^2(A)$ with $\ell^2(\mathbb{Z} \times \{1, \dots, N\})$.

For $\xi < 0 < \eta$, choose orbit sequences $x_k o \rightarrow \xi$ and $y_k o \rightarrow \eta$ tracking rays. For sufficiently large k , the separator identity gives

$$\frac{G(x_k, y_k)}{G(x_k, e)G(e, y_k)} = \left(\frac{G(x_k, \cdot)}{G(x_k, e)} \right)_A M_A \left(\frac{G(\cdot, y_k)}{G(e, y_k)} \right)_A. \quad (6.2)$$

The vectors on the right converge *strongly in* $\ell^2(A)$ to $(\check{K}(g, \xi))_{g \in A}$ and $(K(g, \eta))_{g \in A}$, respectively. Here is the estimate justifying this essential point. The two-sided Green comparison (5.3), followed by the explicit hyperbolic distance formula, bounds the ratios along a ray to a fixed endpoint on one side of the axis by

$$C_{\xi, j} e^{-|n|\tau} \quad \text{or} \quad C_{\eta, j} e^{-|n|\tau}, \quad g = a^n b_j.$$

The constants are uniform in the point along that ray; bounded tracking errors change them by a fixed factor. To see the decay at the endpoint directly, write $b_j i = u_j + iv_j$ and use

$$P_{a^n b_j i}(t) = \frac{e^{n\tau} v_j (1 + t^2)}{(t - e^{n\tau} u_j)^2 + e^{2n\tau} v_j^2} = O_t(e^{-|n|\tau}), \quad t \neq 0, \infty.$$

More explicitly, along a geodesic ray w_t from i to ξ , $d(z_o, w_t) - t$ decreases to $\beta_\xi(z_o, i)$. Hence $\exp(d(i, w_t) - d(z_o, w_t)) \leq P_{z_o}(\xi)$; a bounded error in tracking the ray multiplies this bound by a fixed constant. Applying this to (5.3) gives the uniform bound just used. Its square is summable over A . Coordinatewise Martin convergence and dominated convergence in ℓ^2 now apply. Boundedness of M_A , rather than an unjustified interchange of an entrywise double sum and a boundary limit, yields

$$\Theta(\xi, \eta) = \check{K}(A, \xi)^T M_A K(A, \eta). \quad (6.3)$$

The transpose denotes the real bilinear pairing, extended complex bilinearly when appropriate. All these kernels are real.

LEAN: `StripNaimKernel`; `StripNaimContinuity`; `NaimOrbitPairing`.

Write $f_+(x) = d\nu/dx$ and $f_-(x) = d\check{\nu}/dx$ in the finite boundary chart. In logarithmic coordinates define

$$\begin{aligned} k_g^+(t) &= e^t K(g, e^t) f_+(e^t), \\ k_g^-(t) &= e^t \check{K}(g, -e^t) f_-(-e^t). \end{aligned} \quad (6.4)$$

Thus $k_g^+(t) dt$ is the restriction of $g_*\nu$ to the positive half-line, pulled back by $t \mapsto e^t$. The analogous statement for k_g^- uses $g_*\check{\nu}$ and $t \mapsto -e^t$. In particular,

$$k_{a^n b_j}^\pm(t) = k_{b_j}^\pm(t - n\tau), \quad \int_{\mathbb{R}} k_g^\pm(t) dt \leq 1. \quad (6.5)$$

The density bounds (5.2), applied after pushforward, give $k_{b_j}^\pm(t) \leq C_j e^{-|t|}$. Consequently

$$\sup_t \sum_{n \in \mathbb{Z}} \sum_{j=1}^N k_{a^n b_j}^\pm(t) < \infty. \quad (6.6)$$

Define the analysis operators

$$H_\pm : L^2(\mathbb{R}) \longrightarrow \ell^2(A), \quad (H_\pm \varphi)_g = \int_{\mathbb{R}} \varphi(t) k_g^\pm(t) dt.$$

They are bounded: weighted Cauchy–Schwarz gives

$$\left| \int \varphi k_g^\pm \right|^2 \leq \left(\int |\varphi|^2 k_g^\pm \right) \left(\int k_g^\pm \right) \leq \int |\varphi|^2 k_g^\pm,$$

and summation uses (6.6). The estimate using $\|k_g^\pm\|_\infty$ separately would not provide this summable control over the translates.

In the real chart the Liouville current is, up to normalization, $|\xi - \eta|^{-2} d\xi d\eta$. Thus $J = cJ_0$ and (6.3) give, after absorbing normalization into $c > 0$,

$$c \ell(s - t) = k^-(s)^T M_A k^+(t), \quad \ell(u) = \frac{1}{4 \cosh^2(u/2)}. \quad (6.7)$$

All operators now have specified domains. Convolution by $c\ell$ on $L^2(\mathbb{R})$, denoted $\mathcal{C}$, satisfies

$$\mathcal{C} = H_-^* M_A H_+ : L^2(\mathbb{R}) \xrightarrow{H_+} \ell^2(A) \xrightarrow{M_A} \ell^2(A) \xrightarrow{H_-^*} L^2(\mathbb{R}). \quad (6.8)$$

Indeed, first pair (6.7) against compactly supported bounded test functions in both variables. The bound $\|M_A\| \|k^-(s)\|_2 \|k^+(t)\|_2$ and the preceding periodization estimates justify the vector-valued integrals and their pairings. Then extend by density and boundedness. This avoids treating M_A as an entrywise nonnegative matrix, which it need not be.

LEAN: `KernelFactorization`; `NormalizedTwoSidedSingularity`.

7 The Fourier obstruction

We use $\widehat{f}(\omega) = \int_{\mathbb{R}} f(t) e^{-2\pi i \omega t} dt$, with its unitary extension to L^2 . The following elementary fact is the finite-generator obstruction for a fixed translation lattice; compare Bownik [4].

Lemma 7.1. *For any $h_1, \dots, h_N \in L^2(\mathbb{R})$ and $\tau > 0$, the closed linear span of $\{h_j(\cdot - n\tau) : n \in \mathbb{Z}, 1 \leq j \leq N\}$ is a proper subspace of $L^2(\mathbb{R})$.*

Proof. Set $q = 1/\tau$. For $\omega \in [0, q]$ form the $N \times (N+1)$ matrix

$$B(\omega)_{j,l} = \overline{\widehat{h_j}(\omega + lq)}, \quad 0 \leq l \leq N.$$

Choose measurable representatives of the Fourier transforms. Rank-nullity gives $\dim \ker B(\omega) \geq 1$ for every ω where the entries are finite, hence almost everywhere. We record a measurable choice, since pointwise existence alone would not suffice. For $\varepsilon > 0$ let

$$Q_\varepsilon(\omega) = \varepsilon(B(\omega)^* B(\omega) + \varepsilon I)^{-1}.$$

Its entries are measurable. The spectral theorem for a finite Hermitian matrix shows that $Q_{1/m}(\omega)$ converges, entry by entry, to the orthogonal projection $Q(\omega)$ onto $\ker B(\omega)$: on an eigenvalue $\lambda \geq 0$, the multiplier tends to 1 if $\lambda = 0$ and to 0 otherwise. Thus Q is measurable. Moreover,

$$\sum_{l=0}^N \|Q(\omega) e_l\|^2 = \operatorname{tr} Q(\omega) \geq 1.$$

At least one column has norm at least $(N+1)^{-1/2}$. Take the first such column and normalize it, obtaining a measurable unit vector $v(\omega) \in \ker B(\omega)$.

Define $\widehat{\varphi}(\omega + lq) = v_l(\omega)$ for $0 \leq \omega < q$, $0 \leq l \leq N$, and define it to be zero elsewhere. Then $\|\widehat{\varphi}\|_2^2 = q > 0$. For every n, j , Plancherel gives

$$\langle \varphi, h_j(\cdot - n\tau) \rangle = \int_0^q e^{2\pi i n \tau \omega} \sum_{l=0}^N v_l(\omega) \overline{\widehat{h_j}(\omega + lq)} d\omega = 0.$$

The equality uses $e^{2\pi i n \tau l q} = 1$. All integrals are legitimate by Cauchy–Schwarz. This nonzero φ proves the assertion. $\square$

LEAN: KernelProjection; MeasurableKernel; FrequencyKernel; FourierReduction.

On the other hand, convolution by ℓ has trivial kernel. An explicit calculation compatible with the formal proof is useful. Put $\gamma(t) = \exp(t - e^t)$ and $\tilde{\gamma}(t) = \gamma(-t)$. Substituting $u = e^s$ gives

$$(\gamma * \tilde{\gamma})(t) = e^{-t} \int_0^\infty u e^{-(1+e^{-t})u} du = \frac{e^{-t}}{(1 + e^{-t})^2} = \ell(t).$$

Likewise $\widehat{\gamma}(\omega) = \Gamma(1 - 2\pi i \omega)$, so

$$\widehat{\ell}(\omega) = |\Gamma(1 + 2\pi i \omega)|^2 = \begin{cases} \frac{2\pi^2 \omega}{\sinh(2\pi^2 \omega)}, & \omega \neq 0, \\ 1, & \omega = 0. \end{cases} \quad (7.1)$$

This is everywhere strictly positive. The convolution theorem and Plancherel imply $\ker \mathcal{C} = \{0\}$.

LEAN: LiouvilleFourier.

Apply Lemma 7.1 to $h_j = k_{b_j}^+$. By (6.5), it produces a nonzero $\varphi \in \ker H_+$. Equation (6.8) would then give $\mathcal{C}\varphi = 0$, a contradiction. We have proved:

Theorem 7.2. *Under the hypotheses of Theorem 1.1, if $\Gamma \backslash \mathbb{H}^2$ is compact, then $\nu \perp m_o$.*

LEAN: CocompactHittingSingularity; FuchsianCocompactSingularity.

8 Quasiconvex orbits and a Green-distance criterion

An orbit Γo is *quasiconvex* if some $R < \infty$ satisfies $[xo, yo] \subset N_R(\Gamma o)$ for all $x, y \in \Gamma$. Let $\Lambda \subset \partial\mathbb{H}^2$ denote the orbit limit set.

Lemma 8.1. *If Γo is quasiconvex, either $m_o(\Lambda) = 0$ or $\Gamma \backslash \mathbb{H}^2$ is compact.*

Proof. Suppose $m_o(\Lambda) > 0$. Choose a finite-chart Lebesgue density point ξ of Λ . Along the vertical ray $w_t = \xi + ie^{-t}$, the Poisson approximate identity gives $m_{w_t}(\Lambda^c) \rightarrow 0$. Quasiconvexity, applied to segments with endpoints tending to ξ , provides $g_t \in \Gamma$ at uniformly bounded distance from w_t ; changing the initial point of the ray only changes this bound by a constant. Visual measures at points at distance at most R' have density ratios between $e^{-R'}$ and $e^{R'}$. Invariance of Λ gives

$$m_o(\Lambda^c) = m_{g_t o}(\Lambda^c) \leq e^{R'} m_{w_t}(\Lambda^c) \rightarrow 0.$$

Since Λ is closed and m_o has full support, $\Lambda = \partial\mathbb{H}^2$. Every complete geodesic is then a limit of orbit-endpoint geodesic segments. Quasiconvexity and properness imply that every point on such a geodesic lies within a uniform distance of Γo . Hence the orbit is coarsely dense in $\mathbb{H}^2$. The image of a sufficiently large compact ball covers the quotient, proving compactness. $\square$

LEAN: QuasiconvexConical; QuasiconvexLimitSetDichotomy; QuasiconvexFullBoundary.

Because $\nu(\Lambda) = 1$, Lemma 8.1 and Theorem 7.2 prove singularity for every quasiconvex orbit.

LEAN: FuchsianQuasiconvexSingularity.

Lemma 8.2. *If constants $c, C < \infty$ satisfy $D_G(g) \leq cD(g) + C$ for every g , then Γo is quasiconvex.*

Proof. Let $|g|$ be word length for $S \cup S^{-1}$. Since $r(P) < 1$, choose $q < 1$ and $A < \infty$ with $\|P^n\| \leq Aq^n$. No S -path of length less than $|g|$ reaches g , so

$$G(e, g) \leq \sum_{n \geq |g|} Aq^n = \frac{A}{1-q} q^{|g|}.$$

Thus $D_G(g) \geq \alpha|g| - \beta$ for some $\alpha > 0$. Combining with the hypothesis gives $|g| \leq c_1 D(g) + c_2$. Conversely $D(g) \leq L'|g|$, where $L' = \max_{s \in S \cup S^{-1}} d(o, so)$. By left invariance, the orbit map is a quasi-isometric embedding of the Cayley graph. Connect adjacent orbit points by geodesic segments. Images of word-geodesics are uniform quasi-geodesics; the Morse lemma in $\mathbb{H}^2$ puts the geodesic between their endpoints within a fixed distance of them. Each image edge is itself within L' of its vertices. This proves orbit quasiconvexity. $\square$

LEAN: WordDistance; GreenOrbitQuasiconvex.

9 Dirichlet ends and bounded-height endpoints

The remaining argument constructs the finite separators needed to prove a Green-distance upper bound under hypothetical nonsingularity. It uses no classification of finitely generated Fuchsian groups.

Call $p \in \partial\mathbb{H}^2$ a *bounded-height endpoint* if, for a chart $B \in \mathrm{PSL}_2(\mathbb{R})$ with $B\infty = p$, there is $C_B < \infty$ such that

$$\mathrm{Im}(B^{-1}go) \leq C_B \quad (g \in \Gamma). \quad (9.1)$$

The condition is independent of the chart: two such charts differ by a real affine map fixing ∞ , which multiplies heights by a positive constant. It is also Γ -invariant, by reindexing g .

Lemma 9.1. *If $\Gamma \backslash \mathbb{H}^2$ is noncompact, there exists a bounded-height endpoint.*

Proof. The closed Dirichlet cell about o is

$$\mathcal{D} = \{w \in \mathbb{H}^2 : d(w, o) \leq d(w, go) \text{ for every } g \in \Gamma\}.$$

Every point has a nearest orbit point, since the orbit is proper. Translating by that point puts it in $\mathcal{D}$. If $\mathcal{D}$ were bounded, its compact closure would therefore cover the quotient. Noncompactness supplies $w_n \in \mathcal{D}$ escaping every ball. A subsequence converges in the closed disk compactification to $p \in \partial \mathbb{H}^2$. Choose $B\infty = p$ and write $z = B^{-1}o$, $v = B^{-1}go$, $u_n = B^{-1}w_n$. Then $u_n \rightarrow \infty$ and $d(u_n, z) \leq d(u_n, v)$.

We claim that $\text{Im } v \leq \text{Im } z$. If $\delta = \text{Im } v - \text{Im } z > 0$, the distance inequality, using $\cosh d(u, z) = 1 + |u - z|^2 / (2 \text{Im } u \text{Im } z)$, becomes

$$\delta |u_n|^2 \leq 2A \text{Re } u_n + C, \quad A = (\text{Im } v) \text{Re } z - (\text{Im } z) \text{Re } v, \quad C = (\text{Im } z)|v|^2 - (\text{Im } v)|z|^2.$$

This bounds $|u_n|$ by the quadratic inequality $\delta |u_n|^2 \leq 2|A||u_n| + |C|$, contradicting $u_n \rightarrow \infty$. Thus every g satisfies $\text{Im}(B^{-1}go) \leq \text{Im}(B^{-1}o)$, as required. $\square$

LEAN: ProjectiveDirichletCell; DirichletHeightBound; ProjectiveDirichletEnds.

Lemma 9.2. *For every noncompact quotient, bounded-height endpoints are dense in $\partial \mathbb{H}^2$.*

Proof. Recall that the closure of every boundary orbit contains Λ . For completeness, take a sequence $g_n o \rightarrow \xi \in \Lambda$ and normalized matrix representatives converging to a rank-one matrix. Their boundary actions converge to ξ away from at most one repelling point. Any nonempty closed invariant set contains a point other than that repelling point, since nonelementarity excludes finite orbits. It therefore contains ξ . This proves the assertion about orbit closures.

If $\Lambda = \partial \mathbb{H}^2$, the orbit of the endpoint supplied by Lemma 9.1 is dense, and bounded height is invariant. If $\Lambda \neq \partial \mathbb{H}^2$, its complement is dense: choose a point outside Λ ; its orbit stays outside and its closure contains Λ . Every $p \notin \Lambda$ is a bounded-height endpoint. Indeed, in a chart sending p to infinity, an orbit of unbounded height would have a subsequence converging to infinity, making p an orbit limit point. This proves density also in this case. $\square$

LEAN: BoundedHeightEndpoints; FuchsianDirichletDeficit.

10 Finite separators and harmonic deficits

Take two distinct bounded-height endpoints and send them to 0 and ∞ . The transformed orbit satisfies, for fixed $C, D > 0$,

$$y \leq C, \quad \frac{y}{x^2 + y^2} \leq D.$$

For each $R < \infty$, its intersection with $|x| \leq Ry$ lies in a compact subset of $\mathbb{H}^2$, since

$$\frac{1}{D(1 + R^2)} \leq y \leq C, \quad |x| \leq RC. \tag{10.1}$$

Consequently only finitely many group elements lie in this axis strip. Choose R with $\text{arsinh } R > L$, where L bounds jump length. An edge between opposite sides of the strip cannot avoid it: distance to the axis is $\text{arsinh}(|x|/y)$, and a crossing geodesic has to meet the axis. Enlarging the finite strip by the finitely many one-step predecessors gives a finite

set through which every path exiting the chosen side must pass. In particular, paths whose boundary limit is outside the closure of that side must visit this finite set.

LEAN: `BoundedHeightSeparators`.

Here is the analytic consequence of a finite separator. For a boundary region E , suppose every path started at x and converging in E visits a finite set A . The strong Markov property at first entrance gives, as measures restricted to E ,

$$x_*\nu|_E = \sum_{a \in A} F_A(x, a) a_*\nu|_E.$$

Thus, writing $K_\nu(x, \eta) = d(x_*\nu)/d\nu(\eta)$,

$$K_\nu(x, \eta) = \sum_{a \in A} F_A(x, a) K_\nu(a, \eta) \quad \text{for almost every } \eta \in E. \quad (10.2)$$

First-hit domination bounds $K_\nu(a, \eta) \leq G_0/G(e, a)$. Set

$$C_A = 1 + \sum_{a \in A} \frac{G_0}{G(e, a)G(a, e)}.$$

Then $K_\nu(a, \eta) \leq C_A G(a, e)$, and the Green entrance decomposition implies

$$K_\nu(x, \eta) \leq C_A \sum_{a \in A} F_A(x, a) G(a, e) \leq C_A G(x, e) \quad (\eta \in E \text{ a.e.}). \quad (10.3)$$

For $x = g^{-1}$ this reads

$$D_G(g) - \sigma_\nu(g, \eta) \leq \log C_A + \log G_0. \quad (10.4)$$

There is no infinite sum or boundary-limit exchange in this argument.

LEAN: `FiniteEntranceHarmonicDeficit`; `FuchsianFiniteEntrancePaths`.

Define the harmonic shadow and its inverse image by

$$\mathcal{H}_R(g) = \{\eta : D_G(g) - \sigma_\nu(g, g^{-1}\eta) \leq R\}, \quad T_R(g) = g^{-1}\mathcal{H}_R(g). \quad (10.5)$$

All sets and cocycle formulas may be modified on a single invariant null set; the group is countable. The cocycle bound (2.8) shows that the deficit in this definition is nonnegative.

Proposition 10.1. *If bounded-height endpoints are dense, every sequence (g_n) has a subsequence, still denoted (g_n) , and a finite set $Z \subset \partial\mathbb{H}^2$ with the following property: for every open $U \supset Z$ there exists R such that, eventually,*

$$U^c \subset T_R(g_n) \quad \text{modulo a } \nu\text{-null set.} \quad (10.6)$$

One may take Z empty or a singleton.

Proof. If (g_n) has a constant subsequence, its cocycle is bounded below by (2.8), so take $Z = \emptyset$ and a sufficiently large R . Otherwise pass to a subsequence with $g_n^{-1}o \rightarrow \xi \in \partial\mathbb{H}^2$ and take $Z = \{\xi\}$. Given $U \ni \xi$, choose bounded-height endpoints on either side of ξ so that the closed arc between them containing ξ is contained in U . In the associated chart, $g_n^{-1}o$ is eventually on that side of the axis. The finite-strip construction provides a fixed finite separator for every path from these starting points with boundary limit in U^c . Apply (10.4). $\square$

LEAN: `FuchsianDeficitCompactness`; `FuchsianDirichletDeficit`.

11 Common shadows for harmonic and visual measures

Suppose, towards a contradiction, that $\nu \not\ll m_o$. Then $\nu \ll m_o$ by Proposition 2.1. There is a measurable, exactly invariant set E such that

$$\nu \sim \rho := m_o|_E, \quad 0 < \rho(\partial\mathbb{H}^2) \leq 1. \quad (11.1)$$

Indeed, start with the set where $d\nu/dm_o > 0$ and remove the countable union of its null failures of invariance. Quasi-invariance of both measures makes this removal harmless. On E the logarithmic cocycle of ρ equals the visual cocycle. Set $H = \log(d\nu/d\rho)$; it is finite almost everywhere and satisfies (2.6).

LEAN: FuchsianVisualCarrier.

Conjugate so that $o = i$. For $g \in \Gamma$ put $t = D(g)$ and choose $A_g \in \mathrm{PSL}_2(\mathbb{R})$ with $A_g i = i$ and $A_g(i e^t) = g i$. For $r > 0$ define

$$\mathcal{V}_r(g) = A_g(\partial\mathbb{H}^2 \setminus [-e^t r, e^t r]).$$

For $t = 0$, any such choice suffices. Writing $a_t z = e^t z$, the element $k_g = g^{-1} A_g a_t$ fixes i , and

$$g^{-1} \mathcal{V}_r(g) = k_g(\partial\mathbb{H}^2 \setminus [-r, r]). \quad (11.2)$$

Since k_g preserves m_i ,

$$m_i((g^{-1} \mathcal{V}_r(g))^c) = \frac{2}{\pi} \arctan r \leq \frac{2r}{\pi}. \quad (11.3)$$

The Poisson formula gives, for $\xi \in g^{-1} \mathcal{V}_r(g)$,

$$D(g) - \log(1 + r^{-2}) \leq \sigma_{m_i}(g, \xi) \leq D(g). \quad (11.4)$$

For example, before applying A_g the relevant density at η is $e^t(1 + \eta^2)/(e^{2t} + \eta^2)$, which lies between $e^t/(1 + r^{-2})$ and e^t when $|\eta| > e^t r$. Finally, from every sequence (g_n) we can extract a subsequence with $k_{g_n} \rightarrow k$ in the compact stabilizer of i . For any neighborhood U of $p = k(0)$, choose r small enough; then eventually

$$(g_n^{-1} \mathcal{V}_r(g_n))^c \subset U. \quad (11.5)$$

LEAN: VisualShadows; VisualShadowExceptionalLimit.

Use a countable increasing family of *mixed* shadows:

$$\mathcal{S}_N(g) = \mathcal{H}_N(g) \cap \mathcal{V}_{1/(N+1)}(g), \quad T_N^*(g) = g^{-1} \mathcal{S}_N(g), \quad N \in \mathbb{N}. \quad (11.6)$$

The following two consequences are the precise shadow geometry needed.

Lemma 11.1. *Assume Proposition 10.1. Then:*

1. *There is N_0 such that, for each $N \geq N_0$ and $\lambda \in \{\nu, \rho\}$, constants $c_\lambda > 0$ and $C_\lambda < \infty$ independent of g satisfy*

$$\lambda(T_N^*(g)) \geq c_\lambda, \quad (11.7)$$

$$|\sigma_\lambda(g, \xi) - D_\lambda(g)| \leq C_\lambda \quad (\xi \in T_N^*(g) \text{ a.e.}), \quad (11.8)$$

$$e^{-D_\lambda(g) - C_\lambda} \leq \lambda(\mathcal{S}_N(g)) \leq e^{-D_\lambda(g) + C_\lambda}, \quad (11.9)$$

where

$$D_\nu(g) = D_G(g), \quad D_\rho(g) = D(g).$$

2. From every sequence (g_n) one can extract a subsequence and find an $N \geq N_0$ and finite $F \subset \Gamma$ such that eventually

$$\partial\mathbb{H}^2 = \bigcup_{a \in F} aT_N^*(g_n) \quad \text{modulo a } \nu\text{-null set.} \quad (11.10)$$

Proof. First, harmonic inverse shadows have uniformly almost full ν -mass as $N \rightarrow \infty$. Otherwise choose g_N with $\nu(T_N(g_N)^c) \geq \varepsilon$. Apply deficit compactness to a subsequence. As ν is atomless, choose an open neighborhood U of its finite exceptional set with $\nu(U) < \varepsilon$. For some fixed R and all sufficiently large subsequence indices, $T_R(g_N)^c \subset U$ modulo null sets. Since $N \geq R$ eventually, monotonicity contradicts the choice of g_N . Finite-measure absolute continuity transfers this uniform estimate from ν to ρ . Equation (11.3) and $\nu, \rho \ll m_i$ similarly give uniformly almost full mass for the visual inverse shadows. The intersection therefore has a uniform positive mass for both measures once N is large.

On the mixed inverse shadow, (10.5) and the global upper cocycle bound give $D_G(g) - N \leq \sigma_\nu(g, \xi) \leq D_G(g)$. Equation (11.4) gives the analogous visual estimate. Integrating the Jacobian formula (2.5) over $T_N^*(g)$, using its positive mass and the fact that each measure has total mass at most one, gives (11.9) after increasing the constants. This proves (1).

For (2), combine the finite exceptional set from deficit compactness with the single exceptional point from (11.5); call their union Z . There is a finite $F \subset \Gamma$ with $\bigcap_{a \in F} aZ = \emptyset$. Indeed, a point in the intersection of all translates would have its whole inverse orbit in the finite set Z , contradicting nonelementarity; compactness reduces the empty intersection to finitely many translates. Choose a sufficiently small open neighborhood U of Z so that $\bigcap_{a \in F} aU = \emptyset$. Both exceptional-set statements, and then enlargement of N , give $U^c \subset T_N^*(g_n)$ almost everywhere for all sufficiently large n . Since $\bigcup_{a \in F} aU^c = \partial\mathbb{H}^2$ and translates preserve null sets, (11.10) follows. $\square$

LEAN: FuchsianMixedDeficitUniformMass; FuchsianMixedDeficitEstimates; FuchsianDeficitAECover.

12 From nonsingularity to a uniform Green comparison

We give the measure comparison argument explicitly. This is the step that turns the preceding geometric information into a global bound; it is not an application of an external singularity theorem. The use of common shadows is related to the general Patterson–Sullivan rigidity framework of Kim–Zimmer [5].

Proposition 12.1. *Under (11.1) and the two conclusions of Lemma 11.1, there is a constant $C < \infty$ such that*

$$|D_G(g) - D(g)| \leq C \quad (g \in \Gamma). \quad (12.1)$$

Proof. Choose $M < \infty$ such that the bounded-density window

$$W = \{\xi \in E : |H(\xi)| \leq M\}$$

has positive ν -measure. By (2.2), almost every sample path with limit in W satisfies

$$\nu(Z_n^{-1}W^c) \rightarrow 0. \quad (12.2)$$

Since $\rho \ll \nu$ and both are finite, the same is true with ρ . Fix one such path and put $g_n = Z_n$ along it.

Fix any $N \geq N_0$ and write $S_n = \mathcal{S}_N(g_n)$, $T_n = g_n^{-1}S_n$. By (11.8), (11.7), and the Jacobian formula,

$$\frac{\lambda(S_n \setminus W)}{\lambda(S_n)} \leq \frac{e^{2C_\lambda}}{c_\lambda} \lambda(g_n^{-1}W^c) \rightarrow 0, \quad \lambda \in \{\nu, \rho\}.$$

For all sufficiently large n , each shadow therefore has at least half of its mass in W . On W , $e^{-M}\rho \leq \nu \leq e^M\rho$. It follows that

$$\frac{1}{2}e^{-M} \leq \frac{\nu(S_n)}{\rho(S_n)} \leq 2e^M.$$

Compare this with (11.9). There is $L_N < \infty$ with

$$|D_G(g_n) - D(g_n)| \leq L_N \tag{12.3}$$

for every n , after including the finitely many initial values. The concentration property holds on the chosen path independently of N , so this argument supplies (12.3) for every integer $N \geq N_0$ on that same path.

LEAN: FuchsianShadowPathComparison.

Apply part (2) of Lemma 11.1 to this sequence. Pass to its subsequence and fix the resulting N and finite F . For each $a \in F$, quasi-invariance and (12.2) give

$$\nu(ag_n^{-1}W^c) \rightarrow 0.$$

Pass to a further subsequence so that these error measures are summable, simultaneously for all $a \in F$. Borel–Cantelli implies that, for almost every ξ , eventually

$$g_na^{-1}\xi \in W \quad \text{for every } a \in F.$$

At the same time the finite cover (11.10) supplies an $a \in F$ with $\eta = a^{-1}\xi \in T_N^*(g_n)$. Consequently both $\eta \in T_N^*(g_n)$ and $g_n\eta \in W$ hold. The coboundary identity, the two cocycle estimates, and (12.3) give

$$\begin{aligned} |H(\eta)| &\leq |H(g_n\eta)| + |\sigma_\nu(g_n, \eta) - \sigma_\rho(g_n, \eta)| \\ &\leq M + C_\nu + C_\rho + L_N. \end{aligned}$$

There is also a uniform bound J on $|\sigma_\nu(a, \zeta) - \sigma_\rho(a, \zeta)|$ for $a \in F$ and almost every ζ : use the global Green cocycle bounds (2.8) and $|\sigma_\rho(a, \zeta)| \leq D(a)$. Applying (2.6) to $\xi = a\eta$ yields

$$|H(\xi)| \leq B := M + C_\nu + C_\rho + L_N + J \quad \text{almost everywhere.} \tag{12.4}$$

Thus $e^{-B}\rho \leq \nu \leq e^B\rho$ globally. This return argument is also obtainable from vanishing error measures directly, as in the Lean implementation; the summable subsequence makes the almost-everywhere conclusion particularly transparent.

LEAN: TranslatedConcentrationReturnCover; ReturnCoverDensityRigidity;
FuchsianShadowCorrections.

Finally compare the masses of $\mathcal{S}_N(g)$ for arbitrary g , using (12.4) and (11.9). Their ratio is between e^{-B} and e^B , hence

$$|D_G(g) - D(g)| \leq B + C_\nu + C_\rho.$$

This proves (12.1) for the whole group. □

LEAN: BoundedSequenceShadowRigidity; FuchsianShadowGeometryRigidity.

13 Completion of the proof

Proof of Theorem 1.1. Boundary convergence and the stated properties of its actual law were established in Proposition 2.1. If the quotient is compact, apply Theorem 7.2.

Suppose the quotient is noncompact and, for a contradiction, that $\nu \not\ll m_o$. Then $\nu \ll m_o$, and the carrier construction (11.1) applies. Lemmas 9.1 and 9.2 provide a dense set of bounded-height endpoints. Proposition 10.1 supplies deficit compactness; Lemma 11.1 supplies the common shadows and their finite corrected covers. Proposition 12.1 now yields $D_G(g) \leq D(g) + C$. Lemma 8.2 makes the orbit quasiconvex. By Lemma 8.1, either $m_o(\Lambda) = 0$ or the quotient is compact. The first alternative contradicts $\nu \ll m_o$ and $\nu(\Lambda) = 1$; the second contradicts noncompactness. Thus $\nu \perp m_o$ in all cases.

On $\mathbb{R}$, the density of m_o is strictly positive relative to Lebesgue measure. The hitting measure has no atom at ∞ . The same conclusion therefore holds relative to Lebesgue measure in the finite boundary chart. $\square$

LEAN: `FuchsianSingularity.fuchsian_hittingMeasure_singular_lebesgue`.

The formal final theorem splits the noncompact argument into full and proper limit sets. The exposition combined them in Lemma 9.2; the ingredients and resulting separator estimate are the same. In particular, nonuniform lattices and infinite-covolume groups both fall within the theorem. Semigroup generation, finite support, discreteness, and nonelementarity are retained throughout; symmetry and finite covolume are never added.

A How to locate the proof in Lean

The entry point is `Singularity/FuchsianSingularity.lean`. Its principal declaration is

`Singularity.fuchsian_hittingMeasure_singular`.

The group parameter is a subgroup of $\mathrm{PSL}(2, \mathbb{R})$ with discrete topology. The law is specified by a finite set s , positive weights on s , total mass one, and `Submonoid.closure(s) =  $\mathbb{T}$` . The hitting measure is the law of the constructed geometric boundary limit, not an abstract measure assumed to have shadow estimates. The companion declaration `fuchsian_randomWalk_converges_and_hittingMeasure_singular` includes almost-sure convergence.

The following table lists source modules relative to `formalization/Singularity/`; each name has extension `.lean`. It is a navigation guide, not a list of additional hypotheses of the main theorem.

Mathematical step	Main source modules
Walk and actual limit	<code>FuchsianHittingMeasure</code> ; <code>FuchsianBoundaryConcentration</code> ; <code>FuchsianHittingErgodicity</code>
Green compression	<code>Green</code> ; <code>Operators</code> ; <code>GreenSeparator</code>
Martin–Naïm package	<code>GeometricNaimKernel</code> ; <code>GeometricNaimContinuity</code> ; <code>GeometricNaimCurrent</code> ; <code>GeometricNaimQuotient</code>
One-sided comparison	<code>OneSidedNaimRigidity</code> ; <code>NaimCurrentRigidity</code> ; <code>GreenVisualUpper</code> ; <code>GreenVisualLower</code>
Strip factorization	<code>StripNaimKernel</code> ; <code>StripNaimContinuity</code> ; <code>KernelFactorization</code> ; <code>NormalizedTwoSidedSingularity</code>
Fourier obstruction	<code>KernelProjection</code> ; <code>MeasurableKernel</code> ; <code>FrequencyKernel</code> ; <code>FourierReduction</code> ; <code>LiouvilleFourier</code>

Mathematical step	Main source modules
Cocompact conclusion	CocompactHittingSingularity; FuchsianCocompactSingularity
Quasiconvex reduction	GreenOrbitQuasiconvex; QuasiconvexLimitSetDichotomy; QuasiconvexFullBoundary; FuchsianQuasiconvexSingularity
Dirichlet end	ProjectiveDirichletCell; DirichletHeightBound; ProjectiveDirichletEnds; BoundedHeightEndpoints
Finite separators	BoundedHeightSeparators; FiniteEntranceHarmonicDeficit; FuchsianFiniteEntrancePaths; FuchsianDirichletDeficit
Common shadows	FuchsianMixedDeficitShadows; FuchsianMixedDeficitUniformMass; FuchsianMixedDeficitEstimates; FuchsianDeficitAECover
Density rigidity	FuchsianShadowPathComparison; TranslatedConcentrationReturnCover; ReturnCoverDensityRigidity; FuchsianShadowGeometryRigidity
Final assembly	FuchsianSingularityFromDeficitCompactness; FuchsianSingularity

The recorded project verification reports a successful build of 4471 jobs and an audit of 2591 declarations. The allowed foundational axioms are `propext`, `Classical.choice`, and `Quot.sound`. This describes the saved verification report; producing this exposition does not itself rerun that complete audit. The pinned toolchain is `leanprover/lean4:v4.34.0-rc2`, with `mathlib` commit `3649549a1e4b19461e912299ca7127d8831b79fa`.

From the project root, with Lean’s `elan` toolchain manager installed, the usual reproduction commands are

```
lake exe cache get
lake build
```

The source distribution includes its toolchain and dependency lock files. The large `.lake` directory is a regenerable dependency/build cache. For the mathematical source of this document, run `pdflatex` twice on `singularity-from-lean.tex`; the bibliography is embedded.

B Scope of this exposition and classical inputs

This document proves the singularity conclusion by the argument organized in the Lean development, while stating its foundational boundary package as Proposition 4.1. It is not a line-by-line translation of all supporting formal declarations. Standard results used at that level include the admissible nonamenability spectral criterion, nonelementary boundary convergence, the cocompact Ancona–Martin theory, Hopf ergodicity for the compact hyperbolic geodesic flow, the Morse lemma, Plancherel’s theorem, and the martingale convergence theorem. Their hypotheses and the places where they enter are indicated in the text. The noncompact Dirichlet-end, finite-entrance, mixed-shadow, and density rigidity steps are spelled out rather than replaced by an appeal to a classification or singularity theorem.

A machine-checked source theorem and a human mathematical exposition are different artifacts: the former certifies its formal statement relative to its foundations; the latter must also communicate conventions, dependencies, and mathematical identifications clearly. The source-module references above are provided to make that comparison possible.

References

- [1] W. Woess, *Random Walks on Infinite Graphs and Groups*, Cambridge Tracts in Mathematics **138**, Cambridge University Press, 2000. <https://doi.org/10.1017/CB09780511470967>.
- [2] S. Gouëzel, *Martin boundary of random walks with unbounded jumps in hyperbolic groups*, Annals of Probability **43** (2015), 2374–2404. <https://doi.org/10.1214/14-AOP938>.
- [3] S. Cantrell and R. Tanaka, *Invariant measures of the topological flow and measures at infinity on hyperbolic groups*, Journal of Modern Dynamics **20** (2024), 215–274. <https://doi.org/10.3934/jmd.2024006>.
- [4] M. Bownik, *The structure of shift-invariant subspaces of $L^2(\mathbb{R}^n)$* , Journal of Functional Analysis **177** (2000), 282–309. <https://doi.org/10.1006/jfan.2000.3635>.
- [5] D. M. Kim and A. Zimmer, *Rigidity for Patterson–Sullivan systems with applications to random walks and entropy rigidity*, preprint, arXiv:2505.16556. <https://arxiv.org/abs/2505.16556>.